



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing– II part– Thursday, February 1st, 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please describe how the "Permutation collision resolution" works. Is it related to the parallel gather pattern? (2)

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- B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=0; i<100; i++)
    for (j=0; j<10; j++)
        a[i][j] = f(a[j][i]);
```

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Exercise n. 2

Answer the following questions about OpenMP (without an explanation, the answer will be considered invalid).

A. Which errors will arise compiling the following code snippet? (2)

```
#pragma omp parallel for shared(num_steps) reduction(+:sum) default(none)
for (i=1;i<= num_steps; i++)
{
  x = array[i] * 2;
  sum = sum + x;
}
```

B. Describe the way in which 15 iterations are divided among 10 threads with a static schedule and a chunk size of 2, and with a dynamic schedule with a chunk size of 1. (2)

Exercise n. 3

Answer the following questions about CUDA (without an explanation, the answer will be considered invalid).

- A. The expression to map thread index to data index for a kernel that needs to process an element of a 1D vector is $i = \text{blockIdx.x} * \text{blockDim.x} + \text{threadIdx.x}$. Is this statement true or false? Explain the meaning of the thread hierarchy variables used. (2)

- B. Describe the impact of CUDA atomic operations on the throughput of a program. (1)

- C. Assume that a kernel is launched with 128 thread blocks each of which has 1024 threads. If a variable is declared as a shared memory variable, how many versions of the variable will be created through the lifetime of the execution of the kernel? (1) **This question can be skipped in case of a passed challenge.**

Exercise n. 4

Answer the following questions about parallel programming and Halide (without an explanation, the answer will be considered invalid).

- A. Describe the method to qualitatively evaluate a parallel algorithm through its data flow graph. (2)

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- B. In Halide, it is better to start scheduling the input stage of the pipeline and proceed towards the output. Is this statement true or false? Explain. (2)

[illegible]

